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Historical Biogeography of Spiny Lobsters in the Genus *Panulirus* (Achelata: Palinuridae)

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ABSTRACT

Aim: *Panulirus* is the spiny lobster genus with the most living species, containing 22 recognised species split into two lineages distinguished by habitat preference. Diversification has been proposed to occur due to geographic events affecting the distribution of adults and the dispersal potential of long-lived larvae by oceanic currents. Previously, the most thorough biogeographic inference for *Panulirus* described diversification influenced by vicariance events resulting from plate tectonics and oceanic currents from the Palaeocene to the Pleistocene. We present updated comprehensive phylogenetic hypotheses and historical biogeography reconstruction models for the spiny lobster genus *Panulirus*.

Location: Tropical and subtropical oceanic regions, especially the Indo-West Pacific.

Taxon: The spiny lobster genus *Panulirus* (White 1847); infraorder Achelata, fam. Palinuridae.

Methods: We assembled a dataset of 250 sequences representing mitochondrial (COI and 16S) and nuclear (18S, 28S and H3) gene fragments for 59 members of the Achelata, including 22 living *Panulirus* species and 4 subspecies. Phylogenetic relationships were reconstructed from all concatenated gene fragments and time-calibrated using fossils. Ancestral distributions across deep time were modelled, and informed historical biogeographic inferences were made.

Results: The Achelata and the Palinuridae emerged in the Tethys during the Triassic epoch (~231 and ~225 Mya, accordingly). After the ancestor of *Panulirus* diverged during the late Jurassic (~164 Mya), *Panulirus* L1 and L2 originated in the Western and Indo-Pacific during the early (~135 Mya) and late (~87 Mya) Cretaceous, respectively. The genus *Panulirus* experienced the most diversification during the Eocene (~52 Mya), then continued to diversify until the Pliocene (~3 Mya), with the greatest concentration of ancestral species inhabiting the Indo-West Pacific.

Main Conclusions: Divergence within the Achelata has been primarily driven by shifts in plate tectonics and oceanic current patterns. Dispersal and, secondarily, vicariance were supported as the main driving forces behind *Panulirus* diversification. Geographic events likely significantly impacted the dispersal and settlement of long-lived phyllosoma larvae, thus altering gene flow patterns.

1 | Introduction

In decapod crustaceans, the infraorder Achelata contains the spiny (fam. Palinuridae) and slipper (fam. Scyllaridae) lobsters,

characterised by the enlarged first antennae, a motile phyllosoma larval stage, and the absence of chelae (Holthuis 1946). Within Palinuridae, two main clades have been recognised: Silentes and Stridentes. Representatives of the clade Stridentes

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primarily inhabit calm equatorial regions and possess a stridulating organ for sound production, while the *Silentes* lack a stridulating organ and typically inhabit more turbulent mid-latitude waters (George and Main 1967; George 2006). Members of the family Palinuridae inhabit all major oceans except the Antarctic and Arctic regions, ranging from shallow waters to 1400 m depths in tropical, subtropical and warm temperate habitats (George and Main 1967; Holthuis 1991; Chan 2010).

Panulirus is the spiny lobster genus with the most living species, containing 22 recognised species and 4 subspecies distributed throughout shallow tropical and subtropical oceans (George 2006). Of these species, 14 inhabit the Indo-Pacific and 6 the Atlantic Ocean (George 2006; Giraldez and Smyth 2016). Two *Panulirus* lineages are recognised and distinguished by habitat preference (George and Main 1967; McWilliam 1995; Ptacek et al. 2001; Patek and Oakley 2003). Whereas species belonging to Lineage 1 prefer clear coastal waters, representatives in Lineage 2 frequent shallow, turbid regions sometimes influenced by river runoff (George 2006). In the two lineages, diversification has been proposed to occur due to geographic events influencing vicariance, geographic distribution and the dispersal of long-lived larvae, which might be constrained or favoured by oceanic currents (George 2006). Insight into diversification in the genus *Panulirus* offers a foundation for further clarification of extant taxa relationships and ancestral divergence scenarios.

In marine environments, plate tectonics generally act as a global vicariance mechanism for intertidal and subtidal vertebrate and invertebrate taxa, with dispersal induced by prevailing winds and ocean currents (Crisci 2001; Queiroz 2005). Dispersal dynamics vary according to species or population vagility; thus, the same event may prompt different responses in different species or populations (Kodandaramaiah 2009). Additionally, extinction events provide insights into where dispersal limits exist (Wiens and Donoghue 2004). Previous research into the historical biogeography of marine systems has revealed novel influences on distribution and diversification, including the significance of currents and depth in larval dispersal, habitat colonisation, population isolation, and population connectivity (Queiroz 2005; Singh et al. 2018).

Various historical scenarios have been proposed to explain the diversification and dispersal of spiny lobsters in the genus *Panulirus* (George and Main 1967; Pollock 1990, 1992, 1993; George 1997, 2005, 2006). Among them, the most comprehensive scenario suggested that diversification in the genus *Panulirus* was influenced by geographic change via plate tectonics and oceanic currents, which prompted broad and localised environmental changes contributing to speciation mechanics, including vicariance (George 1997, 2005, 2006). However, the current understanding of the historical biogeography of the genus *Panulirus* is primarily an inference-based verbal model (see Supporting Information, Data S1 for a detailed description).

Understanding the role of dispersal and vicariance events on the divergence of modern-day *Panulirus* spp. and their present distributions might help explain how these ecologically and

economically important spiny lobsters will respond to global climate change. Our study aimed to understand the historical biogeography of the spiny lobster genus *Panulirus* and shed light on how divergence happens in species with high dispersal ability (via teleplanic larvae). We formally tested the comprehensive historical biogeographic scenario proposed by George (1997, 2005, 2006) for *Panulirus* spp. by: first, developing phylogenies based on concatenated nuclear and mitochondrial gene fragments; second, time-calibrating divergence across the phylogeny; and third, producing Bayesian estimates of ancestral geographic distributions. The resulting ultrametric phylogeny including probable ancestral ranges was used to reconstruct dispersal scenarios for historical biogeographic inferences.

2 | Methods

2.1 | Taxon Sampling for the Molecular Dataset

Phylogenetic analyses were conducted using nuclear and mitochondrial gene markers to reveal relationships within the genus *Panulirus* and closely related clades within the infraorder Achelata. These analyses served as the backbone to explore the historical biogeography of the genus *Panulirus*.

For phylogenetic inference, we included 59 species and subspecies within the infraorder Achelata (Tables S2 and S3). The ingroup consisted of 25 species/subspecies in the genus *Panulirus* (20 species, plus 4 currently accepted subspecies and 1 formerly accepted subspecies), as well as 24 other representatives of all genera in the family Palinuridae (*Linuparus*, *Jasus*, *Justitia*, *Nupalirus*, *Palinurus*, *Palinustus*, *Phyllamphion*, *Projasus*, *Puerulus*, and *Sagmariasus*). The outgroup included 10 members of the family Scyllaridae representing all three subfamilies (Arctidinae, Ibacininae, and Thenidae) with species from the genera *Ibacus*, *Scyllaridae*, *Scyllarus*, and *Thenus*.

The genetic dataset was generated from the mitochondrial gene fragments cytochrome oxidase unit 1 (COI) and 16S, plus the nuclear gene fragments 18S, 28S, and histone 3 (H3). These 5 markers were selected because including slowly (nuclear) and rapidly (mitochondrial) evolving genes improves the resolution of deep and shallow relationships during phylogenetic inference (Toon et al. 2009). Additionally, these five gene fragments were informative in previous phylogenetic studies focusing on the Achelata (i.e., Ptacek et al. 2001; Patek and Oakley 2003; Palero et al. 2009, 2016; Bracken-Grissom et al. 2014). In total, 250 sequences were compiled for these five gene fragments in 59 species within the Achelata. We retrieved 180 of these sequences from NCBI's GenBank database and generated 70 sequences from 19 laboratory specimens (15 *Panulirus* species; Tables S2 and S3).

Muscle tissue samples (representing 15 *Panulirus* spp.; Tables S2 and S3) were collected from lobster pereopods and/or abdomens stored in 95% ethanol at the Clemson University Crustacean Collection. Total genomic DNA was extracted from muscle tissue samples using either the G'NOME DNA extraction kit (COI) or the OMEGA BIO-TEK EZNA Blood and Tissue DNA Kit (16S, 18S, 28S, H3), following the manufacturer's protocol.

2.2 | DNA Sequencing

The five gene regions of interest (COI, 16S, 18S, 28S, and H3) were amplified from total genomic DNA via polymerase chain reaction (PCR). For PCR amplification of COI (~650 bp), 16S (~557 bp), 18S (~852 bp), 28S (~1408 bp), and H3 (~328 bp), the primers L-COI490 and H-COI2198 (Folmer et al. 1994), 16L2 (Schubart et al. 2000) and 1472 (Crandall and Fitzpatrick 1996), 18SF and 18SR (Giribet et al. 1996), 28SF and 28SR (Whiting 2002), and H3AF and H3AR (Colgan et al. 1998), were respectively utilised. Standard PCR 25 µL reactions (2.5 µL of 10× Taq buffer, 2 µL of 50 mM MgCl₂, 2.5 µL of 8 mM dNTPs, 1.2 µL each of the two primers (10 mM), 0.625 U Taq, 1.25 µL of 20 mM BSI and 8.625 µL double distilled water) were performed for each gene fragment (see Table S1 for conditions). The 16S, 18S, 28S, and H3 PCR products (≥ 20 ng) were purified with ExoSapIT, then sent to the Clemson University Genomics Institute (CUGI; Clemson University, Clemson, SC, USA) for sequencing with an ABI Big Dye Terminator Mix (Applied Biosystems Inc.) in an ABI Prism 3730×1 Genetic Analyser (Applied Biosystems Automated Sequencer). COI PCR products were purified with Qiaquick PCR purification columns (Qiagen Inc., Valencia, California). Purified COI PCR product (≥ 20 ng) was used as a template for sequencing with the Taq DiDeoxy Terminator Cycle Sequencing Kit (Applied Biosystems Inc., Foster City, California) and run on the Applied Biosystems Model 377 Automated DNA Sequencing System at the Molecular Core Research Facility at Idaho State University. Both forward and reverse strands were sequenced for the five gene fragments of interest, then a consensus sequence (COI, ~622 bp; 16S, ~497 bp; 18S, ~614 bp; 28S, ~726 bp; H3, ~328 bp) produced with Sequencher v.5.4.1 (Gene Codes Corp.) and deposited in GenBank (Table S2).

2.3 | Phylogenetic Analyses Using Concatenated Molecular Markers

To strengthen the detection of real phylogenetic groups, ‘multigene’ phylogenetic analyses were conducted using the concatenated COI, 16S, 18S, 28S, and H3 gene fragment datasets. Each gene fragment dataset was first aligned using MUSCLE (Edgar 2004a, 2004b) in MEGA X (Kumar et al. 2018). Saturation rates for all resolved codon sites were then assessed for each of the datasets for genes encoded by the mitochondrial genome (COI and 16S) via the software DAMBE v.7.3.32 (Xia 2018) using the method proposed by Xia et al. (2003) and Xia and Lemey (2009). If the resulting substitution saturation value (Iss) was greater than the critical substitution saturation value (Iss.c), then that codon position (first, second or third) was considered useless for phylogenetics. If the Iss was smaller than the Iss.c and the difference was supported as statistically significant value (p -value < 0.05), then the codon position was considered valuable with little saturation. For the COI and 16S datasets, the Iss values were smaller than the Iss values and supported as statistically significant for both symmetrical and asymmetrical tree topologies.

Regions with high phylogenetic noise-to-signal in the 16S, 18S, and 28S alignments were identified and removed in the web-server GBlocks v.0.91b (http://phylogeny.lirmm.fr/phylo.cgi/one_task.cgi?task_type=gblocks—Castresana 2000; Talavera

and Castresana 2007). Pruning via GBlocks was unnecessary for the genes COI and H3 since they are protein-coding and rarely contain alignment gaps. Default settings in GBlocks were applied for pruning the 16S fragments. However, less stringent selections are often better suited for shorter alignments (Talavera and Castresana 2007). Since a number of 18S and 28S gene fragments represented partial sequences encompassing differing portions of the complete gene, resulting in variable overlapping regions between species, we applied less stringent pruning settings to 18S and 28S that allowed gap positions within the final blocks and less strict flanking positions. Aligned gene fragment datasets were concatenated using the program Concatenator v.0.2.1 (Price et al. 2010; Dunn 2020; Vences et al. 2021, 2022). Additionally, to explore evolutionary divergence among terminals in the phylogenies, pairwise genetic distances within and among species in the Achelata were computed for the mitochondrial sequence datasets (COI and 16S) using the maximum composite likelihood model (Tamura et al. 2004) in MEGA X. The final gene fragment dataset lengths were 427 bp for COI, 393 bp for 16S, 576 bp for 18S, 586 bp for 28S, and 333 bp for H3.

Maximum likelihood (ML) analyses that allowed for different substitution models across the five gene fragment partitions were conducted in the webserver IQ-Tree v.2.1.2 (<http://iqtree.cibiv.univie.ac.at>—Nguyen et al. 2015; Trifinopoulos et al. 2016; Minh et al. 2020) over 1000 ultrafast bootstrap replicates (Minh et al. 2013). For the ML consensus tree, nodal support was evaluated via bootstrap values (bv) (Hoang et al. 2018). Bayesian inference (BI) analyses with the same partition scheme were performed in MrBayes 3.2.7 (Ronquist and Huelsenbeck 2003; Ronquist et al. 2012) sampling on random starting trees for 4 chains over 6 million Markov chain Monte Carlo (MCMC) generations. Sampling and branch length information were recorded every 100th tree. For the BI consensus tree, node support was estimated using posterior probability values (pv). The resulting molecular ML and BI phylogenies were depicted and rooted in FigTree v1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>).

2.4 | Divergence Time Estimation

Following recommendations in Parham et al. (2012) and suggested calibration points strategies (Porter et al. 2005; Zheng and Wiens 2015), molecular clock age constraints and calibration points were assigned for deep, middle, and shallow nodes using fossil species belonging to the genera *Yunnanopalinura*, *Scyllarides*, *Archaeopalinurus*, and *Panulirus*. Assignments were based on the earliest recorded age estimates for these fossils and allowed for a probability distribution 5% outside of the input ages (Table 1).

Age estimates were assigned to fossils based on information sourced from the literature. The oldest fossilised representative of the Achelata, *Yunnanopalinura schrami* (Table 1), was described from a flattened full-body holotype (LPI-40169) and paratype (LPI-41667) that were well preserved in micritic lithographic limestone of the Luoping Biota (a Lagerstätte), within Member II of the Guanling Formation (Zhang et al. 2008; Hu et al. 2011; Feldmann et al. 2012). The presence of conodonts suggests the Luoping Biota formed during the early middle Triassic (Zhang et al. 2009; Hu et al. 2011). Meanwhile, the

TABLE 1 | Achelata divergence analysis calibrations using the earliest recorded fossils of four ancestral species.

Achelata Fossil Assignment	Species Name	Collection Number	Location	Stratigraphic Level	Age Range (Mya)	Calibration Age Assigned (Mya)	Epoch
Achelata	<i>Yunnanopalinura schrami</i> (Feldmann et al. 2012)	Chengdu Inst. of Geology and Mineral Resources LP-40169	Luoping, China	Luoping Biota, Guanling Formation	247.2–242.0	242.0	Middle Triassic
Scyllaridae	<i>Scyllarides punctatus</i> (Woods 1925)	Muséum National d'Histoire Naturelle FR10183	Folkstone, England	Greensand Formation	125.0–100.5	100.5	Early Cretaceous
Palinuridae	<i>Archaeopalinurus levis</i> (Garassino et al. 1996)	Museo Friulano de Storia Natural MFSN Collection	Forni di Sopra, Italy	Middle Doloma de Forni	215.6–212.0	212.0	Early Middle Triassic
<i>Panulirus</i>	<i>Panulirus destombesi</i> (Garassino and Breton 2010)	Muséum National d'Histoire Naturelle MNHN A29028	Pas-de-Calais, France	Albian Clay, Saint-Pô Formation	100.5–66.0	66.0	Late Cretaceous

oldest fossil belonging to the Scyllaridae, *Scyllarides punctatus* (FR10183), was collected from Greensand sandstone, which formed during the early Cretaceous (Woods 1925). Next, as of 1996, 126 flattened and fragmented specimens (exuviae) have been identified as the oldest fossilised species belonging to the family Palinuridae, *Archaeopalinurus levis* (Table 1) (Garassino et al. 1996). Most of these fossils were recovered from dolomitic rock in the middle Doloma de Forni, which formed during the late Triassic and primarily yields crustacean fossils (Garassino et al. 1996). Lastly, *Panulirus destombesi* (Table 1), the only fossilised species belonging to the genus *Panulirus*, was collected from Albian clay in the Saint Pô Formation, which formed during the late Cretaceous (Garassino and Breton 2010).

The program BEAST v.2.5 (Drummond and Rambaut 2007; Drummond et al. 2012; Suchard et al. 2018; Bouckaert et al. 2019; Dellicour et al. 2021), as implemented in the webserver CIPRES v.3.3 (<https://www.phylo.org/index.php/>; 12/17/2023—Miller et al. 2010) was employed to estimate an ultrametric phylogeny. Time-calibration analysis parameters were first configured in BEAUti2 (Drummond et al. 2012; Bouckaert et al. 2014). Fossil assignments followed the topological constraints recovered by the ML phylogeny and utilised the youngest age estimated for each fossil (see Table 1). The five gene fragment datasets (COI, 16S, 18S, 28S, and H3) were partitioned. Site models were unlinked across these five gene fragment partitions, while clock models and trees were linked. A gamma site model applied an estimated substitution rate and followed the GTR + G model of nucleotide evolution for each gene alignment, which allows for different substitution rates between nucleotide pairs and base frequencies. To reduce gamma prior density on low values, GTR exchangeability rates were adjusted (Heath 2016). The rateAG parameter was set to $\alpha=2.0$ and $\beta=0.5$, while the rateAC, rateAT, rateCG, and rateGT parameters were set to $\alpha=2.0$ and $\beta=0.25$. We implemented the optimised relaxed clock model (ORC), which has increased performance speed and works well with correlated and uncorrelated rates and branch lengths (Douglas et al. 2021). The tree prior followed the Yule model with fossil time calibration points under an exponential prior distribution density mean of 10.0 and was implemented under MCMC (Gilks et al. 1995) sampling conducted using a chain length of 100 million generations, recording every 100 generations, and a pre-burn of 1000.

BEAST v.2.5 uses MCMC sampling to estimate divergence times from the posterior distribution of phylogenetic trees. We ran BEAST v.2.5 twice (200 million generations total), and the estimates of marginal posterior probability distributions were visually checked for convergence in TRACER v.1.7 (Rambaut et al. 2018) to determine whether the MCMC chain length was long enough to sufficiently sample the stationary distribution and thus produce reliable divergence time estimates. The resulting posterior distribution trees were then combined into a single file using LogCombiner (Drummond and Rambaut 2007) with a burnin of 20% (21% total burnin) and states resampled to a frequency of 10,000. Fossils were then removed from the combined tree using the FullToExtantTreeConverter tool in BEAUti2. The resulting tree file was then concatenated using TreeAnnotator (Drummond and Rambaut 2007) with a posterior probability limit of 0 (default) and mean node heights to produce a maximum clade credibility ultrametric tree (MCC). The resulting NEXUS file, containing a

phylogeny with node ages (Mya) and 95% highest probability density (HPD) values, was visualised and plotted on a time scale axis in FigTree v1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>).

2.5 | Reconstruction of Ancestral Geographic Distributions

We reconstructed ancestral geographic distributions to explore the historical biogeographic scenario proposed for *Panulirus* by George (2006). First, ancestral ranges were estimated using six different geographic ranges inhabited by extant taxa: Central Atlantic (A), Southern Atlantic (B), Mediterranean Sea (C), Western Pacific (D), Eastern Pacific (E), and Indo-Pacific (F). Extant species distributions were assigned (see Table S5) based on the central prevailing range according to data sourced from the literature, including Holthuis (1991), George (2005, 2006), and Lavery et al. (2014) and online resources, including the Ocean Biodiversity Information System (OBIS; accessed 4 Dec. 2023). Single outlier occurrences were not included. Assuming that ancestral species could inhabit multiple oceanic regions, the ancestral area maximum was set to 6 for all RASP 4.3 (Yu et al. 2015, 2020) and BioGeoBEARS (Matzke 2013) analyses.

Analyses in RASP4.3 (Yu et al. 2015, 2020) utilised MCC ultrametric trees and geographic distribution constraints to reconstruct ancestral ranges. The R package BioGeoBEARS (Matzke 2013), as implemented in RASP4.3, applied Bayesian likelihood analyses to produce a time-stratified historical biogeographical reconstruction of the Achelata. The input was an MCC ultrametric tree file incorporating the assigned geographic ranges of extant taxa. The output plotted the most likely ancestral state dispersal matrix probabilities vs. state probabilities. Analyses considered the free rate parameters dispersal (d), extinction (e), and vicariance (v) with and without the founder event speciation parameter (+J, also known as the jump parameter) to determine whether ancestral state probabilities match biogeographic stochastic maps (BSMs).

Before conducting ancestral range reconstructions, the divergence model that best fit the data was determined by the BioGeoBEARS model test, which uses nested Likelihood Ratio Tests (LRT) and nested Akaike Information Criteria (AIC) with and without the j parameter, as DEC + J is not statistically comparable to DEC (Ree and Sanmartin 2018). The null hypothesis that two models share the same likelihood was accepted or rejected based on *p*-values (<0.05). Overall, BAYAREALIKE + J fit the dataset best. BAYAREALIKE (likelihood version of BayArea) (Landis et al. 2013; Matzke 2013) tests the importance of cladogenesis and anagenesis on the dataset, allowing for sympatric and range copying events (i.e., an ancestor can inhabit multiple areas), but neither subset sympatry nor vicariance.

3 | Results

3.1 | Molecular Phylogeny

The final concatenated data matrix (COI, 56 sequences; 16S, 53 sequences; 18S, 49 sequences; 28S, 40 sequences; and H3, 52 sequences) included 2314 aligned positions for 59 terminals,

representing 25 species and/or subspecies belonging to the genus *Panulirus* alongside 24 other species belonging to the family Palinuridae (ingroup) and 10 representatives of the family Scyllaridae (outgroup). The ML and BI phylogenetic hypotheses exhibited the same general topology (Figure 1). Node support values were generally higher in the BI tree than in the ML tree. For additional support resolving evolutionary divergence within the Achelata, pairwise genetic distances for COI and 16S are provided in Tables S6 and S7.

The family Palinuridae was fully supported (bv = 100; pv = 1.00) as monophyletic by the ML and BI phylogenetic analyses. The ML and BI tree topology also supported a sister relationship between ‘Stridentes’ and ‘Silentes’ lobsters.

The ‘Silentes’ clade was strongly supported by ML and BI (bv = 90; pv = 0.96), and comprised representatives belonging to the genera *Jasus*, *Sagmariasus*, *Projasus*, and *Phyllamphion*. Both ML and BI analyses fully supported *Phyllamphion elegans* as sister to another fully supported clade containing *Jasus* + *Sagmariasus* + *Projasus* (bv = 100; pv = 1.00). In the latter clade, ML and BI topology supported the clade containing *Jasus* + *Sagmariasus* as a sister group to *Projasus*. Additionally, both trees fully supported the monophyly of separate *Jasus* and *Projasus* clades (bv = 100; pv = 1.00).

Within the moderately supported ‘Stridentes’ clade (bv = 79; pv = 0.61), ML and BI analyses fully supported a clade containing representatives of the genera *Linuparus*, *Justitia*, and *Nupalirus*. *Nupalirus japonicus* had an early branching position (bv = 100; pv = 1.00), being a sister taxon to a strongly supported clade containing *Justitia longimana* and a monophyletic *Linuparus* clade (bv = 90; pv = 0.95). The ML and BI analyses recovered a strongly supported clade (bv = 98; pv = 0.99) containing representatives of the genera *Palinurus*, *Puerulus*, *Palinustus*, *Paibythus*, and *Panulirus*. In the latter clade, there was full support for a monophyletic *Palinurus* clade (bv = 100; pv = 1.00). However, there was poor support for a clade containing the genera *Puerulus* and *Palinustus* (bv = 58; pv = 0.57). Although overall, the clade containing *Puerulus* + *Palinustus* was not fully resolved, *Puerulus velutinus* held an early branching position sister to a clade containing all other representatives belonging to the genera *Puerulus* and *Palinustus* (bv = 82; pv = 1.00). Meanwhile, a clade containing *Panulirus* + *Paibythus* was poorly supported in the ML tree (bv = 64) but moderately supported in the BI tree (pv = 0.81). The genus *Panulirus* was fully supported (bv = 100; pv = 1.00) as a monophyletic clade by the ML and BI analyses (Figure 1). Furthermore, the split of the genus *Panulirus* into two sister clades representing Lineages 1 and 2 *sensu* George (2006) was fully supported. Within Lineage 1, both ML and BI analyses fully supported *Panulirus interruptus* as a sister taxon to all other members of this lineage, which could be further subdivided into three major clades. The first major clade within Lineage 1 suggested by ML and BI tree topology was moderately supported by the ML tree (bv = 71), poorly supported by the BI tree (pv = 0.50), and contained *P. guttatus* with an early branching position, arranged as a sister taxon to a fully supported clade consisting of *P. meripurpuratus* and *P. argus*. The second major clade in Lineage 1 that was fully supported contained *P. echinatus* and *P. penicillatus*, and was recovered as a sister clade to the third major clade in Lineage 1 (bv = 99; pv = 1.00). In the ML

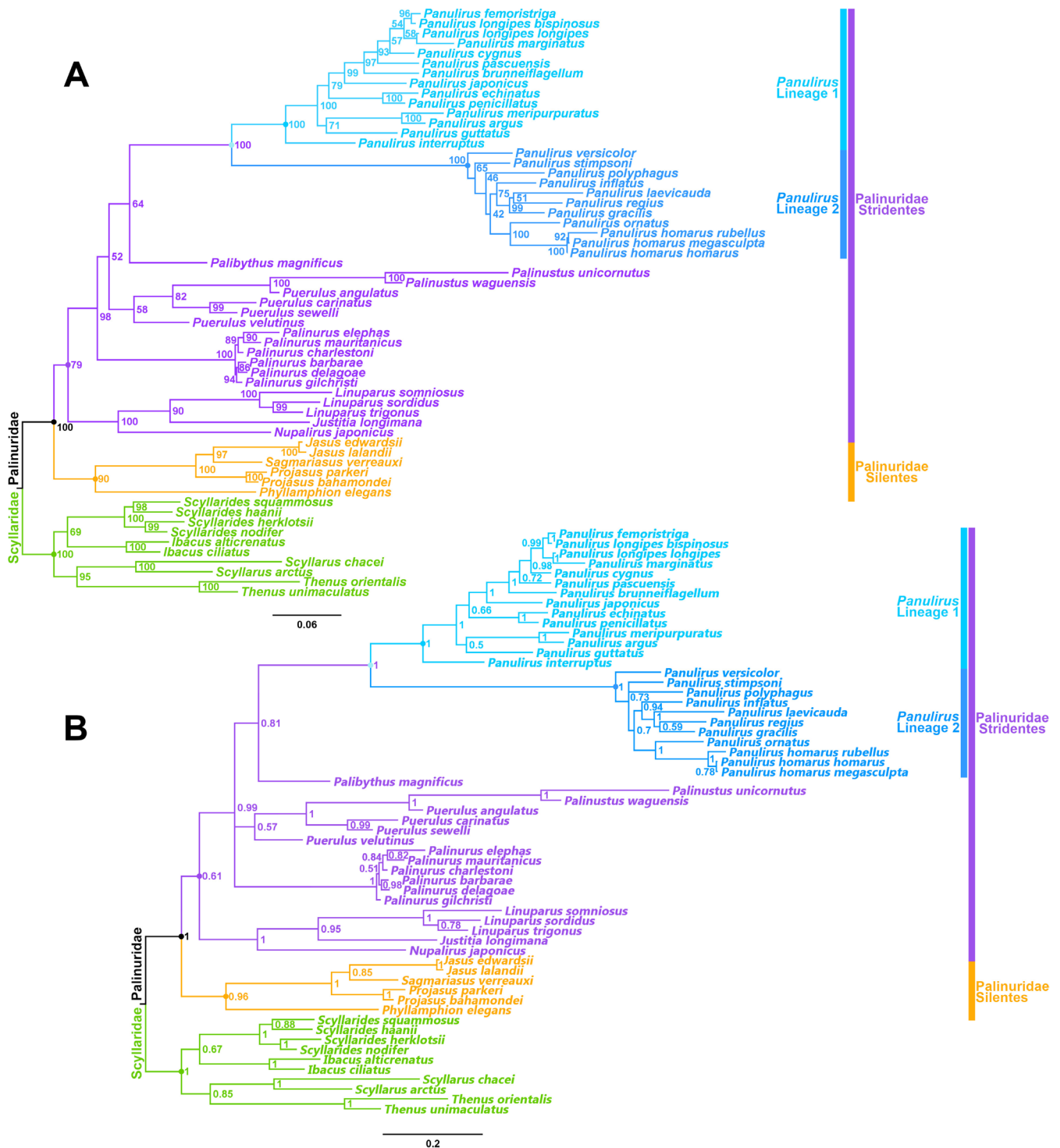


FIGURE 1 | Concatenated gene fragment maximum likelihood (ML) and Bayesian inference (BI) phylogenies of the Achelata. (A) ML tree constructed using the IQ Tree v.2.1.2 webserver. Numbers represent ultrafast bootstrap values produced over 1000 reads. (B) BI tree constructed using MCMC probability distribution sampling in MrBayes 3.2.7. Numbers represent the posterior probability consensus for node arrangement.

and BI trees, the third major clade within Lineage 1 contained *P. japonicus*, *P. brunneiflagellum*, *P. pascuensis*, and *P. cygnus*, in addition to a clade containing an unresolved relationship between *P. femoristriga* + *P. l. bispinosus* (bv=96; pv=1.00) and *P. l. longipes* + *P. marginatus* (bv=58; pv=1.00).

In Lineage 2, the ML and BI trees supported *P. versicolor* at an early branching position (bv=100; pv=1.00). *Panulirus*

versicolor was also arranged as sister to a clade (bv=65; pv=0.73) containing *P. stimpsoni*, *P. polyphagus*, and two major clades consisting of the remaining species belonging to Lineage 2 (bv=75; pv=0.94 and bv=100; pv=1.00, respectively). The relationships between *P. stimpsoni*, *P. polyphagus*, *P. versicolor*, and the two clades representing the remaining Lineage 2 species were not well resolved. The first of these two major clades within Lineage 2 was moderately to strongly supported and

consisted of *P. inflatus*, *P. laevicauda*, *P. regius*, and *P. gracilis* (bv = 75; pv = 0.94). *Panulirus inflatus* held an early branching position (bv = 75; pv = 0.94) in the latter clade and was arranged as sister taxon to a strong to fully supported clade containing *P. laevicauda*, *P. regius*, and *P. gracilis* (bv = 99; pv = 1.00). In the ML tree, *P. gracilis* was a sister taxon to a poorly supported clade containing *P. laevicauda* and *P. regius* (bv = 51). In the BI tree, *P. laevicauda* was a sister taxon to an unresolved clade containing *P. regius* and *P. gracilis* (pv = 0.59). Within the second major Lineage 2 clade, *P. ornatus* was a sister taxon to a fully supported clade containing the *P. homarus* subspecies (bv = 100; pv = 1.00). In the ML tree, *P. homarus homarus* was arranged as sister to a strongly supported clade (bv = 92) containing *P. homarus rubellus* and the currently unaccepted subspecies *P. homarus megasculpta* (superseded by *P. h. homarus*). In the BI tree, *P. h. rubellus* was arranged as sister to a moderately supported *P. h. homarus* + *P. homarus megasculpta* clade (pv = 0.78).

3.2 | Divergence Time Estimation

An ultrametric time-calibrated phylogeny utilising four previously well-described fossil specimens provided divergence time approximations within the Achelata (Figure 2). According to this calibration, the common ancestor of the Achelatan families Palinuridae and Scyllaridae diverged during the Triassic (mean, ~231 Mya; 95% HPD, 212.62–253.89 Mya). The Scyllaridae emerged during the Jurassic (mean, ~170 Mya; 95% HPD, 136.83–206.20 Mya).

The Palinuridae likely branched in the Triassic (mean 225 Mya; 95% HPD, 204.19–248.91 Mya), then split into Silentes (mean, ~180 Mya; 95% HPD, 144.37–206.20 Mya) and Stridentes (mean, ~214 Mya; 95% HPD, 193.53–239.92 Mya). By the mid-Jurassic, the genera representing the ‘Stridentes’ in this phylogeny were divided into three major lineages: the approximately 170 Mya old (95% HPD, 139.10–202.82 Mya) ancestral lineage of *Nupalirus*, *Justitia*, and *Linuparus*; the approximately 169 Mya old (95% HPD, 140.65–197.85 Mya) ancestral lineage of *Puerulus*, *Palibythus*, *Palinustus*, and *Palinurus*; and the approximately 164 Mya old (95% HPD, 143.53–188.75 Mya) *Panulirus* lineage.

The genus *Panulirus* started to diverge in the Cretaceous period with the radiation of Lineage 1 about 135 Mya (95% HPD, 113.49–156.89 Mya) and Lineage 2 about 87 Mya (95% HPD, 74.54–98.65 Mya) (Figure 2). Within Lineages 1 and 2, the most recent ancestral species of modern extant taxa diverged between the Cretaceous (145–66 Mya) and the Pliocene (5–1.8 Mya). The latest Lineage 1 divergence occurred during the Miocene, about 11 Mya (95% HPD, 5.51–16.39 Mya) and represented the ancestor of *P. l. bispinosus* and *P. femoristriga*. The latest Lineage 2 divergence occurred during the Pliocene, about 3 Mya (95% HPD, 0.58–5.90 Mya) and represented the ancestor of *P. h. homarus* and *P. h. megasculpta*.

3.3 | Reconstruction of Ancestral Geographic Distributions

Ancestral ranges in the Achelata were reconstructed using the best-fitting BioGeoBEARS model, BAYAREALIKE+J

(w[AIC_c] = 1.00; $p < 4.2e-09$; Table S4). The RASP4.3 analyses indicated that diversification in the Achelata was influenced by a dispersal event in the Western and Indo-Pacific oceanic regions during the Triassic epoch, about 231 Mya (Figure 3; node 117; 98.19% of all modelled ancestral areas). Subsequently, dispersal events in the Western and Indo-Pacific led to the origin and posterior diversification of the family Palinuridae during the late Triassic, about 225 Mya (node 107; 98.77%) and the Scyllaridae during the mid-Jurassic, about 170 Mya (node 116; 97.18%). Within the Palinuridae, the ancestor of the ‘Stridentes’ underwent further dispersal events in the Western and Indo-Pacific during the late Triassic, about 214 Mya (node 101; 99.85%), while the ‘Silentes’ might have diverged in the Indo-Pacific during the Jurassic, about 180 Mya (node 106; 68.23%).

The diversification of *Panulirus* stemmed from a dispersal event during the Jurassic epoch, approximately 164 Mya (node 83; 98.59%), with an ancestor distributed across the Western and Indo-Pacific regions. From this locale, the Lineage 1 most recent common ancestor might have undergone dispersal and vicariance events in the Western and Indo-Pacific during the early Cretaceous, about 135 Mya (node 72; 57.26%). Meanwhile, the Lineage 2 most recent common ancestor likely dispersed to the Western and Indo-Pacific during the late Cretaceous, about 87 Mya (node 82; 99.83%). Further ancestral range reconstructions for species within the genus *Panulirus* are outlined in Figure 3. A more detailed description for the historical biogeography of the genus *Panulirus* that synthesises the results from the divergence time estimation and the ancestral range reconstruction is provided in the discussion.

4 | Discussion

4.1 | Phylogenetics of the Achelata and Genus *Panulirus*

This study included data from representatives of all genera in the family Palinuridae and all species within the genus *Panulirus* to produce the most comprehensive phylogenetic hypotheses to date for spiny lobsters. Mitochondrial (COI and 16S) and nuclear (18S, 28S and H3) gene fragments were used to generate ML and BI phylogenetic reconstructions of the Achelata that largely concurred with other hypotheses in the literature (Ptacek et al. 2001; Patek and Oakley 2003; Palero et al. 2009; Bracken-Grissom et al. 2014; Davis et al. 2015). Previous phylogenetic inferences have utilised either morphological data, short mitochondrial fragments, a combination of mitochondrial and nuclear markers, or a combination of morphological and molecular data. In line with what was reported before by Ptacek et al. (2001), Patek and Oakley (2003), Palero et al. (2009), Bracken-Grissom et al. (2014), and Davis et al. (2015), this study supported the families Scyllaridae and Palinuridae as monophyletic. Within the Palinuridae, the monophyletic status of the Stridentes and Silentes groups was also supported. This study primarily focused on *Panulirus*, a monophyletic genus within the Stridentes family Palinuridae. Within the genus *Panulirus*, we recovered two fully supported clades previously designated as Lineage 1 and Lineage 2 by Ptacek et al. (2001), Patek and Oakley (2003), and George (2006). However, there were major

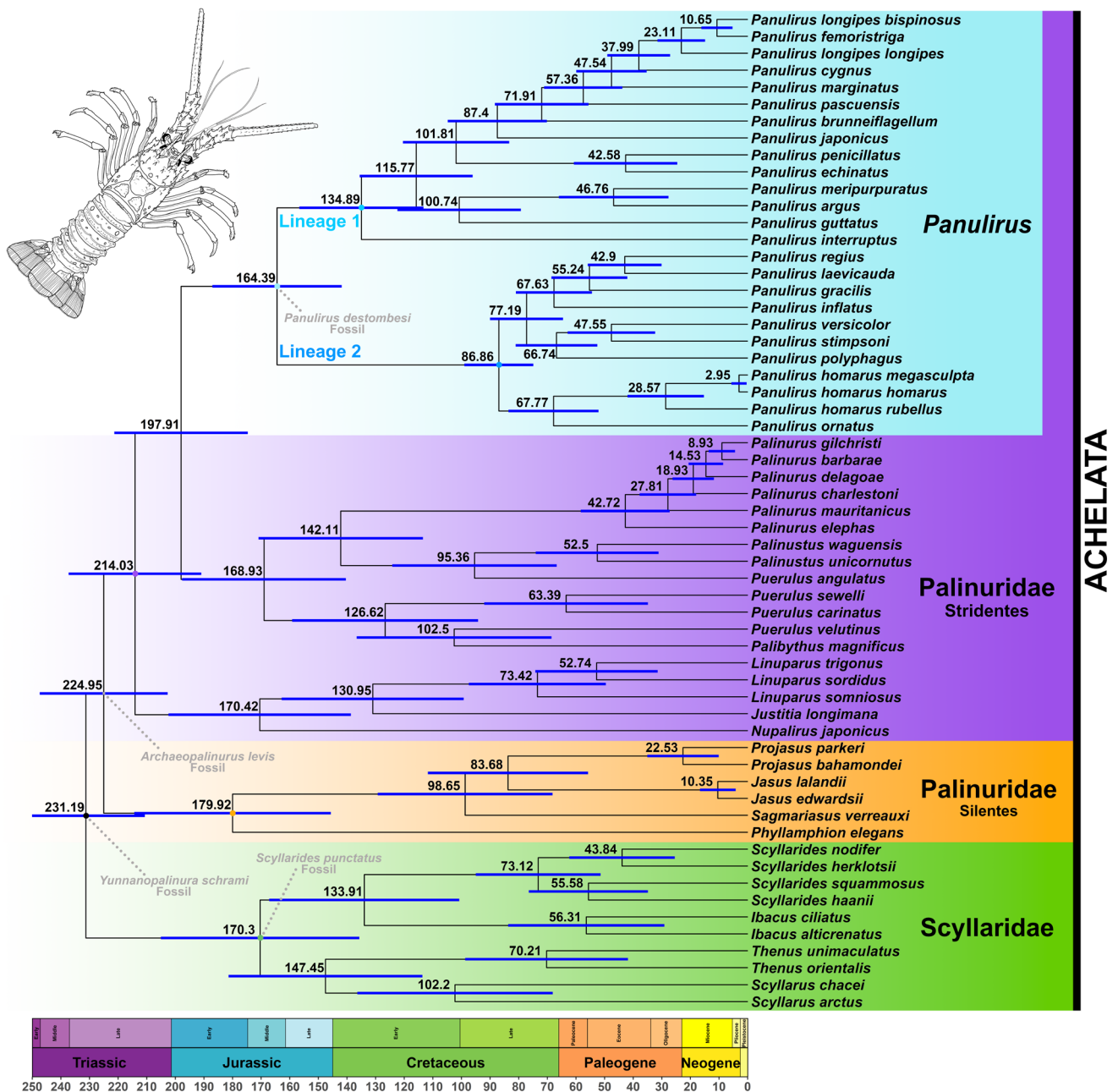


FIGURE 2 | Time-calibrated ultrametric phylogeny of the Achelata reconstructed based on gene fragments. Divergence times were calibrated in BEAST v.2.5 using four fossil calibration points. Nodes represent median divergence times and include 95% HPD bars. Nodes indicating MRCA for highlighted clades of interest are marked with corresponding colours. The geological timescale was configured using the R package ggtree (Yu et al. 2017, 2018; Yu 2020; Xu et al. 2022).

differences between the phylogenies produced herein and prevailing trends in previous phylogenetic reconstructions (Ptacek et al. 2001; Patek and Oakley 2003; Palero et al. 2009; Bracken-Grissom et al. 2014; Davis et al. 2015).

One significant dissimilarity between this and previous studies included the position of *P. argus* and *P. interruptus* in Lineage 1 within the genus *Panulirus*. With the addition of data for *P. meripurpuratus*, *P. argus* no longer held the earliest branching position in the Lineage 1 clade, as in Ptacek et al. (2001), Patek and Oakley (2003) and Davis et al. (2015). Instead, the reconstructions supported *P. interruptus* as the most basal species in

Lineage 1, and *P. argus* was placed in a more derived clade with *P. guttatus* and *P. meripurpuratus*.

Our phylogenetic reconstructions support two significant taxonomic revisions within the genus *Panulirus*, although both instances still require further evidence. First, this study suggests raising *P. h. rubellus* to the species level. While *P. h. homarus* and *P. h. megasculpta* form a well-supported clade with short branch lengths, *P. h. rubellus* had a branch length comparable to the evolutionary divergence between other species within the genus *Panulirus* (Figure 1), pairwise genetic distances between the *P. homarus* subspecies were 0 for the COI and 16S datasets.

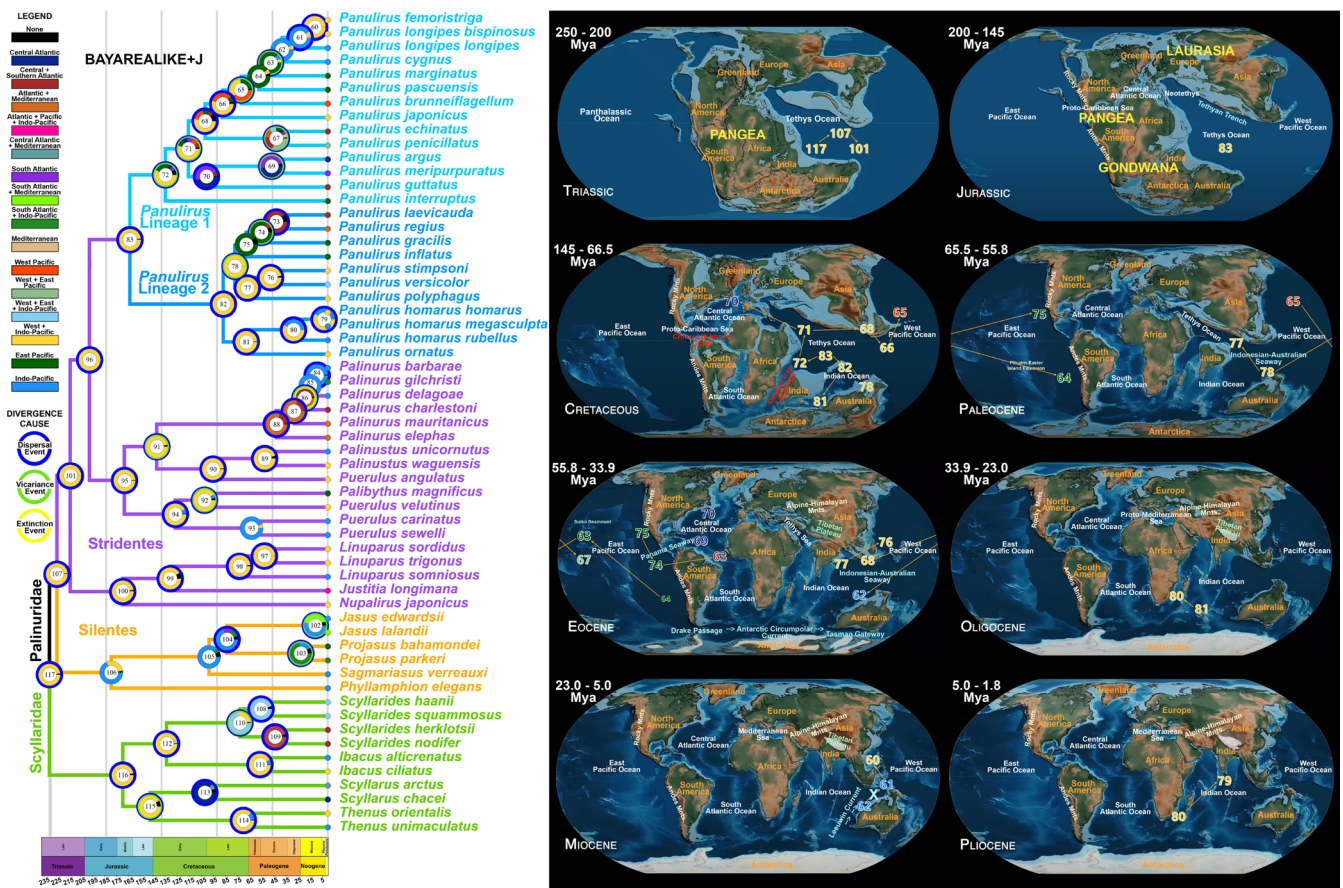


FIGURE 3 | Ancestral geographic distributions and divergence throughout the Achelata reconstructed from an ultrametric phylogeny based on gene fragments. Divergence times were calibrated in BEAST v.2.5 using four fossils. Ancestral ranges were reconstructed from 6 possible areas using RASP4.3 and the BioGeoBEARS model BAYAREALIKE+J and plotted along a geological timescale configured in the R package ggtree (Yu et al. 2017, 2018; Yu 2020; Xu et al. 2022). Maps of plate tectonics and paleogeography were sourced from Scotese (2019).

Meanwhile, the time-calibrated phylogeny indicated a substantial ~26 Mya difference between the divergence times of *P. h. rubellus* and the *P. h. homarus* + *P. h. megasculpta* clade (Figure 2). However, a previous population genetics study (Lavery et al. 2014) utilising mitochondrial (COI, 16S and control region) and nuclear (18S and ITS-1) markers for numerous individuals representing each putative *P. homarus* subspecies and form suggested that *P. h. homarus* and *P. h. megasculpta* lineages were not genetically distinct, and thus *P. h. megasculpta* is not valid. Lavery et al. (2014) also declared that *P. h. rubellus* was not fully reproductively isolated from the *P. h. homarus* populations, and although these lineages were genetically distinct, the extent of hybridisation requires further study. In summation, whether *P. h. rubellus* should be elevated to species status deserves further investigation.

We also suggest reconsideration of the subspecies designation for *P. l. longipes* and *P. l. bispinosus*. Both ML and BI gene tree topology and branch lengths go against the currently accepted nomenclature, which would be expected to confirm a distinct *P. longipes* lineage with a sister relationship between the subspecies *P. l. bispinosus* and *P. l. longipes* (Figure 1). However, the phylogenetic analyses instead support a sister relationship between *P. femoristriga* and *P. l. bispinosus*. Additionally, the branch lengths of *P. l. bispinosus* and *P. l. longipes* were similar to those belonging to currently accepted species within other Achelata

genera, namely *Palinurus* and *Jasus*. A taxonomic revision of the *P. longipes* subspecies was also supported by the time-calibrated phylogeny, which indicated a ~12 Mya difference between the *P. l. bispinosus* and *P. l. longipes* divergence times (Figure 2). However, considering their morphological similarities, it is possible that misidentification of *P. l. longipes*, *P. l. bispinosus* or *P. femoristriga* samples could have occurred for sequences utilised herein. Detailed population genetics studies, such as that conducted by Lavery et al. (2014), are recommended to further evaluate the validity of *P. l. longipes* and *P. l. bispinosus*.

4.2 | Historical Biogeography of the Genus *Panulirus*

Herein, we provide the most comprehensive historical biogeography reconstruction to date for species belonging to the spiny lobster genus *Panulirus* after its initial divergence following the Tethyan Trench formation during the late Jurassic (~164 Mya). The Tethyan Trench potentially acted as a barrier, driving divergence between shallow water species in the then-flooded Tibetan Plateau region and deeper water species in the main Tethys Ocean. Phylogenetic reconstruction coupled with ancestral state analyses of biogeographical regions suggested that cladogenetic events within *Panulirus* primarily corresponded to dispersal and vicariance events stemming from major global

geographic changes from the Cretaceous to the Pleistocene (145–0.01 Mya). These results support the prevalent impact of regional and long-distance dispersal on divergence processes throughout the evolutionary history of the Achelata. However, we do acknowledge that for all historical biogeographic hypotheses, coincidences between geographic and divergence events do not necessarily confirm causality and that there might be other divergence scenarios that are more difficult to estimate for ancient or extinct species without robust palaeoecological data.

By the early Cretaceous (145–100 Mya), the northward separation of India and Madagascar from Antarctica and western Australia expanded the Tethys (Ali and Aitchison 2008; Scotese 2019, 2021) and, consequently, produced mid-ocean ridges with shallower depths (Larson and Pitman III 1972; Seton et al. 2009). About 135 Mya, these rapidly growing ridges likely constrained the dispersal of the most recent common ancestor of *Panulirus* Lineage 1 to clear, deep waters in the eastern Tethys and western Pacific. Following subsequent dispersal across the Pacific, *P. interruptus* likely diversified from the common Lineage 1 ancestral species. As the length of these ridges remained fairly constant throughout the mid to late Cretaceous (Seton et al. 2009), shallow areas along ridges in Indian and western Pacific oceans and within the narrowing northern Tethys also presumably drove the adaptive radiation and vicariance processes responsible for the ancestor of all other species belonging to Lineage 1 about 116 Mya.

By the late Cretaceous (100.5–65.5 Mya), rifting between North America and Eurasia expanded the northern Atlantic and proto-Caribbean Sea, and the position of the African-Arabian landmass diverted equatorial waters northward, creating a sharper climatic gradient north of the Tethys (Gordon 1973; Scotese 2019, 2021). This expansion of new warm, shallow habitat might have facilitated the dispersal of a new ancestor of all Lineage 1 species (excluding *P. interruptus*) from the northern Tethys to the proto-Caribbean Sea, where the ancestor of *P. guttatus* speciated about 101 Mya. From the populations that remained in the eastern Tethys and western Pacific, an ancestor of 12+ Lineage 1 species (excluding *P. interruptus*, *P. guttatus*, *P. argus* and *P. meripurpuratus*) diverged about 102 Mya. Following the rapid seafloor spreading that caused India and Madagascar to continue moving northward (Scotese 2019, 2021), a series of islands formed within shallow seas along continental landmasses and subduction zones in the southwestern Pacific (Zinsmeister 1982; Scotese 2021). As a result, about 87 Mya, the species representing the most recent common ancestor of Lineage 2 likely diverged from the original *Panulirus* ancestral species after dispersing to new warm, shallow, turbid coastal regions along these islands, the India and Madagascar landmass, and southern Europe. Concurrently, the ancestor of *P. japonicus* dispersed in the Western and Indo-Pacific. Later, about 72 Mya, the formation of these new shallow regions likely also triggered dispersal and vicariance events that limited the ancestor of *P. brunneiflagellum* to deeper waters in the western Pacific. Meanwhile, about 77 Mya, an ancestor of 7+ Lineage 2 species (namely *P. laevicauda*, *P. regius*, *P. gracilis*, *P. inflatus*, *P. stimpsoni*, *P. versicolor*, and *P. polyphagus*) likely emerged in the Western and Indo-Pacific; considering the preference of Lineage 2 species for shallow waters with suspended sediment (George and Main 1967; Ptacek et al. 2001; George 2006), this ancestor

was likely isolated in these shallow regions following an initial dispersal event.

Towards the end of the Cretaceous, two major events potentially reduced global temperature (Schulte et al. 2010; Keller 2014). In North America, the impact of the Chicxulub asteroid about 66 Mya might have caused a sunlight-blocking cloud that reduced global temperature and triggered the K-T mass extinction (Schulte et al. 2010). More probably, the multi-phased eruptions of the Deccan Traps lava flats along the eastern Indian coast between 67.4 and 65 Mya likely released volcanic gasses that further contributed to global cooling (Schulte et al. 2010; Keller 2014). Three ancestral species diverged within 1 Mya of these events. The common ancestor of *P. ornatus* and *P. homarus* likely diversified 68 Mya following dispersal to the relatively warmer, shallow equatorial regions in the western and Indo-Pacific. Concurrently, the distribution of the *P. inflatus* ancestor in the eastern Pacific was likely limited to warm equatorial waters. The third species, representing the *P. polyphagus* ancestor, likely diverged about 67 Mya also following a dispersal event within the west and Indo-Pacific.

The Palaeocene (65.5–55.8 Mya) was characterised by a warming climate with tropical equatorial seas (Savin 1977; Lawver and Gahagan 2003; Inglis et al. 2020). This warming climate was primarily caused by thousands of petagrams of carbon releasing into the ocean–atmosphere system over 20 thousand years during the Palaeocene-Eocene Thermal Maximum (McInerney and Wing 2011). In the east, plate subduction along the western North and South American coasts continued to uplift the Rocky and Andes Mountains (Blakey and Ranney 2018; Boschman 2021). The warm climate and resulting rains increased freshwater mountain runoff into the Eastern Pacific, creating new regions of lowered salinity and increased turbidity (McInerney and Wing 2011) that potentially prompted the diversification of the *P. pascuensis* ancestor about 57 Mya.

The Eocene (55.8–33.9 Mya) was a period of numerous plate tectonic shifts (Scotese 2019, 2021). In the Atlantic Ocean, North America and Greenland split from Europe and shifted west, resulting in a plate subduction that caused the continued uplift of the Rocky Mountains along the western North America coast (Blakey and Ranney 2018; Scotese 2021). Plate collisions in Europe also formed the Alpine-Himalayan mountain ranges (Scotese 2021). Notably, about 50 Mya, the collision of the Indian and Asian plates that formed the Himalayas and Tibetan Plateau occurred at the most rapid pace ever estimated, about 15–20 cm per year (Chatterjee et al. 2013; Jagoutz et al. 2015; Scotese 2019, 2021). The continued uplift of numerous mountain ranges likely further intensified freshwater runoff and created turbid regions of lowered salinity along the western North American and Eurasian coasts (McInerney and Wing 2011; Scotese 2021). By about 55 Mya, the ancestor of *P. gracilis* likely diversified in the eastern Pacific and experienced variable salinity conditions as it dispersed along the western North and South American coasts following the currents of the East Pacific Barrier, which circulated from the poles towards the equator, then along the equator to the western Pacific (Cowman and Bellwood 2013).

In the mid-Eocene (47.8–38.0 Mya), about 45 Mya, Australia separated from Antarctica, connecting the Indian and South

Pacific oceans to form the Tasman Gateway (Exon et al. 2002; Lawver and Gahagan 2003; Huber et al. 2004; Sarkar et al. 2019). Meanwhile, the northward shift of South America opened the Drake Passage (Lawver and Gahagan 2003; Scher and Martin 2004, 2006; Sarkar et al. 2019). The opening of the Tasman Gateway and Drake Passage contributed to the formation of the Antarctic Circumpolar Current, which limited heat transport and promoted cold-water upwelling in the southern hemisphere (Exon et al. 2002; Huber et al. 2004; Sarkar et al. 2019). Although the Eocene epoch started with a global warming event (about 30°C) that likely encouraged the initial radiation of Indo-Pacific species throughout warm waters (Zachos et al. 2008), the new Antarctic Circumpolar Current initiated a period of global cooling (below 5°C) and climate seasonality (Miller et al. 1991; Bralower et al. 1995; Zachos et al. 2008; Sarkar et al. 2019). We suggest that these changes in climate and current patterns might have restricted the radiation of numerous ancestral *Panulirus* species in the Pacific. The ancestor of *P. marginatus* emerged about 48 Mya and potentially dispersed throughout equatorial waters of the east Pacific during the period of warming, then became isolated to the warmer region around the Suiko Seamount as the surrounding waters cooled. Later, about 43 Mya, the ancestor of *P. echinatus* and *P. penicillatus* diverged after dispersing from the western to eastern Pacific, then became isolated to warmer shallow or equatorial regions across this range. This divergence pattern would have also allowed for the ensuing split between *P. echinatus* in the Atlantic and *P. penicillatus* in the east Pacific during a more recent epoch.

Meanwhile, in the central Atlantic, dispersal and vicariance events resulted in the divergence of the ancestor of *P. argus* and *P. meripurpuratus* about 47 Mya. Additionally, continued uplift of the Himalayas increased freshwater drainage and sedimentation from the Mekong River, producing a new turbid region in the west and Indo-Pacific (Talbot and Stevenson 1991; Harrison et al. 1992), which likely influenced the dispersal and speciation of *P. stimpsoni* and *P. versicolor's* ancestor about 48 Mya. By about 43 Mya, the common ancestor of *P. regius* and *P. laevicauda* likely travelled across the Panama Seaway before dispersing throughout the central and southern Atlantic Ocean. By the late Eocene (38.0–33.9 Mya), the ancestor of *P. cygnus* potentially emerged about 38 Mya in the Indo-Pacific in response to the northward shift of Australia, changing current patterns and narrowing the Indonesian–Australian Seaway (Scotese et al. 2021).

By the Oligocene (33.9–23.0 Mya), the collision of the African and Eurasian plates closed the Tethys Sea and formed the proto-Mediterranean Sea (Torfstein and Steinberg 2020). With the loss of east–west equatorial current flow, the Earth abruptly began to cool, and the global sea level decreased as the first glaciers formed in Antarctica (Cavelier et al. 1981; Berggren and Prothero 1992; Scotese 2021). Later, after South America and Antarctica further separated, the Antarctic Circumpolar Current contributed to additional southern Pacific cooling (Zinsmeister 1982; Barker and Thomas 2004; Lyle et al. 2007; Martinson 2012). The dispersal of *P. homarus*, the most recent ancestor of *P. h. rubellus*, during the late Oligocene (~29 Mya) was likely restricted to the remaining warm, shallow, turbid coastal habitat that formed across the western and Indo-Pacific due to the fragmentation of the Tethys.

During the Miocene (23.0–5.0 Mya), the collision of the Australian and Asian plates closed the Indonesian–Australian Seaway and fragmented the Pacific–Indian Ocean Throughflow along the Australian coast (Hall 2002; Kuhnt et al. 2004; Scotese 2019, 2021). Remnants of this throughflow along the southwestern coast of Australia formed the Leeuwin Current (George 1997), and its warm waters and eastward flow against prevailing winds prevented cold-water upwelling and reduced heat transfer from the equatorial Pacific to the Indian Ocean (Godfrey and Weaver 1991; Podder et al. 2024). The initial reduction of the Pacific–Indian Throughflow and subsequent formation of the Leeuwin Current barrier likely facilitated new shallow habitat (Wilson and Rosen 1998) and altered the dispersal patterns of the *P. l. longipes* ancestor in the Indo-Pacific during the early Miocene (~23 Mya). By the mid-Miocene, new current patterns likely allowed larvae to disperse and settle along the eastern coast of Asia, resulting in the radiation of the *P. l. bispinosus* and *P. femoristriga* ancestor throughout the western and Indo-Pacific about 11 Mya.

During the Pliocene (5.0–1.8 Mya), the global climate continued to cool and generally became more arid (Savin 1977; Raymo et al. 1996; Haywood et al. 2009). About 5.96 Mya, the Messinian salinity crisis—a period where the Mediterranean almost entirely dried up and most Indo-Pacific species in this region became extinct—resulted in a reduction in the number of species inhabiting the Mediterranean today (Krijgsman et al. 1999). Numerous instances of mountain uplift also occurred, including the continued uplift of the Alps and the Himalayas via the collision of India with Asia (Powell and Conaghan 1973; Trifonov et al. 2012). Erosion of these mountain formations due to intensified Asian monsoon systems likely created regions of increased freshwater runoff with lowered salinity and increased turbidity (Huntington et al. 2006). These regional changes might have impacted the continued divergence of *P. homarus* about 3 Mya and led to the speciation of the ancestor of *P. h. homarus* and *P. h. megasculpta* in the western and Indo-Pacific.

4.3 | Outlook on the Historical Biogeography of the Genus *Panulirus*

Our historical biogeographic analysis partially fits the biogeographic scenario proposed by George (1997, 2005, 2006) for the Achelata and, specifically, the genus *Panulirus*. Overall, our study agreed that divergence within the Achelata likely resulted from a combination of radiation and vicariance events prompted by the emergence of new islands and changes to climate and current systems (George 2006). Within the Palinuridae, this study supported the genus *Phyllamphion* as the earliest offshoot of the Silentes (Davie 1990; Patek and Oakley 2003; George 2006), but our proposed divergence time of ~180 Mya during the Jurassic epoch was earlier than the previous suggestion by George (2006) that divergence occurred during the Pliocene due to the closure of the Panama Seaway.

Our study also indicated that *Panulirus* (~164 Mya) diverged following the formation of the Tethyan Trench barrier during the late Jurassic and is thus not the most recently diverged genus within the Achelata, which contradicts numerous prior studies (George and Main 1967; Pollock 1992; Baisre 1994; McWilliam 1995;

George 2006). Instead, the model suggested that *Jasus* (~10 Mya) might be the most recently diverged genus within Palinuridae, albeit this divergence time was based on gene fragments for only two of the five *Jasus* spp. Additionally, our model suggested that the ancestor of *Panulirus* Lineage 1 diverged during the early Cretaceous epoch, about 135 Mya, after the formation of shallow ridges in the Indo-West Pacific. Following rapid seafloor spreading, the ancestor of *Panulirus* Lineage 2 diverged about 87 Mya after dispersing to new warm, shallow, turbid coastal regions along islands, the India + Madagascar landmass, and southern Europe. Generally, most ancestral species within *Panulirus* diverged earlier in the model than the epochs proposed by George (1997, 2005, 2006) and others (George and Main 1967; Pollock 1992, 1993; McWilliam 1995; Ptacek et al. 2001; Patek and Oakley 2003). This difference likely stems from including a recently discovered *Panulirus* fossil, *P. destombesi* (Garassino and Breton 2010) in the phylogeny time-calibration. The fossil belonging to *P. destombesi* formed during the Late Cretaceous epoch and is approximately 100.5–66.0 Mya, significantly older than the previous estimate that a *Panulirus* ancestor emerged during the Miocene (23.0–5.0 Mya).

So far, only one other study has evaluated divergence times within the Achelata using BEAST and fossil calibrations. Bracken-Grissom et al. (2014) found that the Achelata diverged about 255 Mya, similar to our suggested divergence time of 231 Mya. Although Bracken-Grissom et al. (2014) suggested a substantially earlier divergence time for the Scyllaridae (~217 Mya vs. ~170 Mya, herein), the proposed divergence time for the Palinuridae was similar to our results (~233 Mya vs. ~224 Mya, herein). The same divergence time for the genus *Panulirus* (~164 Mya) was recovered by our study and Bracken-Grissom et al. (2014) despite different fossil selections towards the tree tips.

4.4 | The Role of a Highly Dispersive Larvae in Driving Diversification in the Genus *Panulirus*

Other than vicariance driven by the formation of hard geographical barriers, diversification in the Achelata might be primarily facilitated or constrained by the presence of the highly dispersive planktonic phyllosoma larval stage. Previous studies on decapod larvae have described mark-time moulting—moulting not accompanied by a stage change—in phyllosoma in *Panulirus japonicus* and other Achelata (Williamson 1982; Gore 1985; Booth 1994; Matsuda et al. 2012). This trait is likely useful for extending larval duration in species that disperse over vast distances in environments with a variable food supply (Pollock 1995). Importantly, besides some mixing of phyllosoma and puerulus characters (Polz 1995; Pasini and Garassino 2009; Haug et al. 2013) and a more gradual developmental pattern than that observed in modern representatives (Haug and Haug 2013), fossil phyllosoma dated up to the upper Jurassic have retained laterally orientated appendages (Haug et al. 2013) and undergone little morphological change in the last 200 million years (Polz 1971). This morphological stasis suggests that ancient phyllosoma were also planktonic (Haug et al. 2013) and thus the dispersal potential of ancient spiny lobsters was likely influenced by environmental factors similar to those experienced by modern taxa.

For extant taxa, currents are the main mechanism of both phyllosoma dispersal and retention to natal sites (Sponaugle et al. 2002; George 2006). Essentially, in palinurid larval species for which larval recruitment dynamics have been evaluated, using currents for dispersal increases the chance that larvae radiate to newly established habitats and maintain gene flow connectivity between distant populations (Phillips and Booth 1994; Anger 2006 and references therein). Therefore, lineages with wide geographic ranges and highly dispersive larvae across their evolutionary history are likely readily influenced by geographic changes that interrupt or alter dispersal patterns for significant periods of time. This study indicated that diversification happens in species that exhibit high dispersal ability via teleplanic larvae, which were expected to exhibit greater connectivity among distantly located populations compared to species with direct development (i.e., hymenosomatid crabs, *Neorhynchoplax bovis*—Qhaji et al. 2015; isopods, *Exosphaeroma hylecoetes*; and cumaceans, *Iphinoe truncata*—Teske et al. 2007) or short larval periods (i.e., caridean shrimp, *Thor amboinensis*—Titus et al. 2018; and stomatopods, *Haptosquilla pulchella*—Barber et al. 2002). We argue in favour of future comparative biogeographic studies to understand the effect of larval dispersal ability on diversification (drivers and constraints). For example, how did island species with narrow geographic distributions, such as *P. marginatus* (Hawaii) and *P. pascuensis* (Rapa Nui), diverge much earlier than the islands they currently occupy? Additionally, because distant populations often experience disparate local selective pressures, future studies are also needed to clarify the role of local adaptation in species other than spiny lobsters with wide geographic ranges and high dispersal potential.

4.5 | Historical Impacts on the Divergence of Marine Invertebrates With Highly Dispersive Larvae

The results of this study are also fairly consistent with the mechanisms behind the historical biogeography scenarios inferred for other marine invertebrates with a highly dispersive planktonic larval stage (compared to species with direct development), including palaemonid shrimp (*Pontonia* spp.—Fransen 2002; Frolová et al. 2022), spider crabs (*Maja* spp.—Sotelo et al. 2009), snails (*Echinolittorina* spp.—Williams and Reid 2004; *Nerita* spp.—Frey and Vermeij 2008), limpets (*Nacella* spp.—González-Wevar et al. 2010), urchins (*Diadema* spp.—Lessios et al. 2001), brittle stars (Ophiuroidea—Bribiesca-Contreras et al. 2019), and staghorn corals (*Acropora* spp.—Wallace and Rosen 2006). In agreement with the biogeography scenario for the Achelata and, specifically, the genus *Panulirus*, these prior studies have also suggested that divergence was primarily due to dispersal and/or vicariance events caused by tectonic plate movements, prevailing current patterns and shifting climate. As in our study, for caridean shrimps in the genus *Pontonia*, the fragmentation and closure of the Tethys Sea during the Oligocene or early Miocene epochs likely caused Indo-West Pacific and Atlantic Ocean populations to diverge, and only later did taxa radiate to the East Pacific (Lessios et al. 2001; Fransen 2002; Williams and Reid 2004; Wallace and Rosen 2006; Frey and Vermeij 2008; Sotelo et al. 2009; Bribiesca-Contreras et al. 2019; Frolová et al. 2022). Additionally, prior studies in caridean shrimps, water fleas, brachyuran crabs, snails, limpets, and brittle stars support the more frequent divergence of marine taxa with planktonic larvae during the Eocene to early

Miocene compared to other epochs (Frey and Vermeij 2008; Sotelo et al. 2009; González-Wevar et al. 2010; Xu et al. 2011; Bribiesca-Contreras et al. 2019; Frolová et al. 2022). Further comparisons evaluating the historical biogeography of marine taxa with different planktonic larval periods might help to elucidate conditions impacting larvae dispersal during the Eocene and Miocene epochs.

5 | Conclusions

We present herein the most comprehensive phylogenetic hypotheses and historical biogeography reconstruction to date for the spiny lobster genus *Panulirus*. Phylogenetic hypotheses and historical biogeography reconstructions for the infraorder Achelata highlight how divergence within the spiny lobsters from the Triassic to Pliocene epochs was driven by the opening and closure of seaways, current fragmentation and formation, and fluctuations in climate and salinity. Within the Achelata, historical biogeography models suggested that dispersal and vicariance events drove *Panulirus* speciation probably by influencing the transport of long-lived phyllosoma larvae. Additional studies on phylogenetic relationships and the historical biogeography of the infraorder Achelata should consider two factors. First, genetic differences and disparate divergence times in our models support raising *P. l. longipes*, *P. l. bispinosus*, and *P. h. rubellus* to the species level, although future in-depth population genetics studies are necessary to clarify the validity of these species. Lastly, we recommend modelling divergence times and ancestral areas again as new fossil information becomes available.

Author Contributions

A.M.B. and J.A.B. conceived the ideas with support from M.J.C.; M.J.C. coordinated the sample donation to the Clemson University Crustacean Collection; A.M.B. analysed the data with input from J.A.B.; and A.M.B. headed the writing with support from J.A.B. and M.J.C.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

These sequence data produced herein have been submitted to the GenBank database (ncbi.nlm.nih.gov/genbank) under accession numbers PQ225956–PQ225967 and PQ232563 for COI, PQ223091–PQ223105 for 16S, PQ2223076–PQ223090 for 18S, PQ226094–PQ226108 for 28S and PQ220225–PQ220240 for H3.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** A brief summary of the verbal model for the diversification and dispersal of *Panulirus* spp. over deep time, as proposed by George and Main (1967) and George (1997; 2005; 2006). **Table S1:** PCR cycle conditions for each of the five gene fragments (COI, 16S, 18S, 28S and H3) sequenced in this study. For each gene and cycle, the annealing stage was conducted at 48°C–57°C for 1 min depending on the species. **Table S2:** List of the species and gene fragment data utilised by this study and available on GenBank. **Table S3:** Collection, species identification and location details for the specimens collected and sequenced in this study. Unknown information is marked with ‘UNKN’. **Table S4:** Results of the BioGeoBEARS model test for the divergence model that would be the most plausible time-stratified reconstruction of the ancestral ranges of the Achelata. BAYAREALIKE+J (w[AICc]=1.00 $p < 4.2e-09$) was defined as the most suitable model. **Table S5:** Geographic range assignments for the 59 extant species utilised in the reconstruction of ancestral species distributions using BioGeoBEARS in RASP4.3. The area codes are as follows: Central Atlantic (A), Southern Atlantic (B), Mediterranean Sea (C), Western Pacific (D), Eastern Pacific (E) and Indo-Pacific (F). **Table S6:** Estimates of evolutionary divergence between 16S sequences in the Achelata. **Table S7:** Estimates of evolutionary divergence between COI sequences in the Achelata.